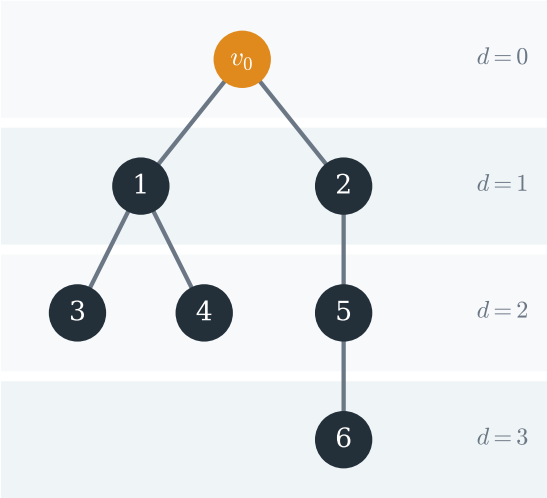


a spanning tree, rooted; key = (distance, vertex)



	$\{0, 1\}$ e_1	$\{0, 2\}$ e_2	$\{1, 3\}$ e_3	$\{1, 4\}$ e_4	$\{2, 5\}$ e_5	$\{5, 6\}$ e_6
1	1	0	-1	-1	0	0
2	0	1	0	0	-1	0
3	0	0	1	0	0	0
4	0	0	0	1	0	0
5	0	0	0	0	1	-1
6	0	0	0	0	0	1

rows: vertices by key $\uparrow$ · columns: their parent edges
zeros below the diagonal, ± 1 on it $\Rightarrow \det = \pm 1$